

LAMPIRAN

HELPER SCRIPTS

Pipeline Generator Scripts untuk EKG Classification Project

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Dokumen ini berisi 4 Python helper scripts:

- `build_workflow_diagram_en.py` — Generate visualization workflow diagram (9 stages)
- `build_extra_viz_cells.py` — Inject precision-recall, calibration, DCA cells ke notebook
- `build_pseudocode_doc.py` — Create Word doc dengan 5 algorithm pseudocode blocks
- `build_extra_viz_doc.py` — Create Word doc dengan 4 evaluation visualization + explanations

1. build_workflow_diagram_en.py

```
# -*- coding: utf-8 -*-
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from matplotlib.patches import FancyBboxPatch, FancyArrowPatch

OUT = r"D:/PROGRAM/ekg_cnn_transformers_/FOTO DOCUMENT/study_workflow_diagram_en.png"

STAGES = [
    ("1. DATA INGESTION", [
        "580 ECG images (JPG/JPEG)",
        "4 folders: Abnormal/Normal x Training/Testing",
        "Source: photographed/scanned printed ECG strips",
    ], "#BFD4F4", "#2E75B6", "white"),
    ("2. VALIDATION & EDA", [
        "File readability validation (cv2.imread)",
        "Class distribution: 204 Abnormal : 376 Normal",
        "Image size & RGB color-bias analysis",
    ], "#FCE9C6", "#C98B1F", "white"),
    ("3. PREPROCESSING", [
        "Grayscale conversion (remove color bias)",
        "CLAHE clip=2.0, grid=8x8 (enhance waveform)",
        "Resize to 224x224 (INTER_AREA)",
        "Normalization to [0, 1] range, float32",
    ], "#D6ECD2", "#4C8C3F", "white"),
    ("4. AUGMENTATION (training data only)", [
        "Conservative: rotation/shift/zoom +-5%",
        "Brightness, mild Gaussian noise & blur",
        "Class-specific: Abnormal 1.5x more aggressive",
    ], "#F6D4D9", "#C4425A", "white"),
    ("5. DATA SPLIT STRATEGY", [
        "Combine all 580 images",
        "5-Fold Stratified Cross-Validation",
        "Class proportion preserved in every fold",
    ], "#E3D7F2", "#7B57A6", "white"),
    ("6. MODELING (2 architectures)", [
        "CNN Baseline: ResNet18 transfer learning",
        "Hybrid: ResNet18 backbone + Transformer Encoder",
        "Backbone partially frozen (layer4 trainable)",
    ], "#E3D7F2", "#7B57A6", "white"),
    ("7. TRAINING", [
        "Adam optimizer, lr=1e-4, 12 epochs, batch=16",
        "CrossEntropyLoss + class-weight (imbalance)",
        "Identical setting for both models (no tuning)",
    ], "#C9EAE6", "#1E8F7D", "white"),
    ("8. EVALUATION", [
        "Accuracy, Precision, Recall, F1, AUC-ROC",
        "Aggregated 5-fold confusion matrix",
        "Training curves & ROC curve",
    ], "#D6ECD2", "#4C8C3F", "white"),
    ("9. OUTPUT & RECOMMENDATIONS", [
        "CNN vs Hybrid performance comparison",
        "Imbalance-handling experiment analysis",
    ], "#D6ECD2", "#4C8C3F", "white"),
]
```

```

        "Recommendations for next steps",
    ], "#DCDCDC", "#555555", "#1a1a1a"),
]

# --- layout constants (data units) ---
TITLE_TOP_PAD = 0.55      # distance from box top to title baseline
FIRST_LINE_PAD = 1.20     # distance from box top to first bullet baseline
LINE_SPACING = 0.60       # spacing between bullet baselines
BOTTOM_PAD = 0.55         # clearance below the last bullet baseline
GAP = 0.55                # vertical gap between boxes
X0, W = 0.6, 8.8
TITLE_BAND = 1.6          # space reserved at top of figure for the main title

def box_height(n_lines):
    return FIRST_LINE_PAD + (n_lines - 1) * LINE_SPACING + BOTTOM_PAD

heights = [box_height(len(lines)) for _, lines, *_ in STAGES]
total_boxes_height = sum(heights) + GAP * (len(STAGES) - 1)
TOP_Y = total_boxes_height + 0.3
FIG_TOTAL_H = TOP_Y + TITLE_BAND

fig, ax = plt.subplots(figsize=(9.5, FIG_TOTAL_H * 0.52))
ax.set_xlim(0, 10)
ax.set_ylim(0, FIG_TOTAL_H)
ax.axis("off")

def box(x, y, w, h, title, lines, facecolor, edgecolor, title_color):
    fb = FancyBboxPatch((x, y), w, h, boxstyle="round,pad=0.08,rounding_size=0.15",
                        linewidth=1.4, edgecolor=edgecolor, facecolor=facecolor, zorder=2)
    ax.add_patch(fb)
    ax.text(x + 0.25, y + h - TITLE_TOP_PAD, title, fontsize=12.5, fontweight="bold",
            color=title_color, va="center", ha="left", zorder=3)
    ty = y + h - FIRST_LINE_PAD
    for line in lines:
        ax.text(x + 0.35, ty, "• " + line, fontsize=9.7, color="#1a1a1a",
                va="center", ha="left", zorder=3)
        ty -= LINE_SPACING

def arrow(x, y1, y2):
    ar = FancyArrowPatch((x, y1), (x, y2), arrowstyle="->", mutation_scale=18,
                        linewidth=1.6, color="#555555", zorder=1)
    ax.add_patch(ar)

y_cursor = TOP_Y
box_edges = []
for (title, lines, fc, ec, tc), h in zip(STAGES, heights):
    top = y_cursor
    bottom = y_cursor - h
    box(X0, bottom, W, h, title, lines, fc, ec, tc)
    box_edges.append((top, bottom))
    y_cursor = bottom - GAP

for i in range(len(box_edges) - 1):
    arrow(X0 + W / 2, box_edges[i][1], box_edges[i + 1][0])

ax.text(X0 + W / 2, TOP_Y + 1.0,
        "Study Workflow: ECG Image Classification\n(CNN Baseline vs Hybrid CNN+Transformer)",
        fontsize=13.5, fontweight="bold", color="#1B3A5C", ha="center", va="center")

plt.subplots_adjust(left=0.02, right=0.98, top=0.99, bottom=0.01)
plt.savefig(OUT, dpi=200, facecolor="white")
print("Saved:", OUT)

```


2. build_extra_viz_cells.py

```
import json, uuid

NB = r"D:\PROGRAM\ekg_cnn_transformers\EKG_CNN_TRANSFORM.ipynb"
nb = json.load(open(NB, encoding="utf-8"))

def md(src):
    return {"cell_type": "markdown", "id": uuid.uuid4().hex[:8], "metadata": {}, "source": src.splitlines(keepends=True)}

def code(src):
    return {"cell_type": "code", "id": uuid.uuid4().hex[:8], "metadata": {}, "execution_count": None, "outputs": [], "source": src}

cells = []
cells.append(md("""---
# Bagian III – Visualisasi Evaluasi Tambahan

Tiga visualisasi tambahan untuk memperdalam analisis performa CNN Baseline vs Hybrid CNN+Transformer, melengkapi Bagian 7 (

cells.append(md("""## 12. Precision vs Recall per Fold (5-Fold CV)

Line chart yang menghubungkan titik precision-recall tiap fold secara berurutan (Fold 1 -> 5), untuk memvisualisasikan l

cells.append(code('''# --- Grafik 5: Precision (y) vs Recall (x) per Fold, garis mengikuti urutan fold ---
fig, ax = plt.subplots(figsize=(7.5, 6.5))

for dfm, label, color, marker in [
    (cnn_metrics_df, "CNN Baseline", "#339af0", "o"),
    (hybrid_metrics_df, "CNN+Transformer Hybrid", "#e8590c", "s"),
]:
    dfm_sorted = dfm.sort_values("fold")
    ax.plot(dfm_sorted["recall"], dfm_sorted["precision"], marker=marker, linestyle="--",
            color=color, label=label, markersize=9, linewidth=1.8)
    for _, row in dfm_sorted.iterrows():
        ax.annotate(f"F{int(row['fold'])}", (row["recall"], row["precision"]),
                    textcoords="offset points", xytext=(6, 6), fontsize=8.5, color=color)

ax.set_xlabel("Recall (Abnormal)")
ax.set_ylabel("Precision (Abnormal)")
ax.set_title("Precision vs Recall per Fold (5-Fold Cross-Validation)\n\nGaris menghubungkan Fold 1 -> Fold 5 secara berur

ax.legend(loc="lower left")
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print("Interpretasi: titik yang berkerumun di kiri-atas (precision tinggi, recall rendah) menunjukkan")
print("model konservatif; titik yang lebih ke kanan menunjukkan model lebih berani mendeteksi Abnormal."'''))

cells.append(md("""## 13. Calibration Curve (Reliability Diagram)

Mengukur apakah probabilitas yang dikeluarkan model bisa dipercaya secara harfiah: sumbu-X = probabilitas prediksi rata-rata

cells.append(code('''# --- Grafik 6: Calibration Curve (Reliability Diagram) ---
from sklearn.calibration import calibration_curve
from sklearn.metrics import brier_score_loss

frac_pos_cnn, mean_pred_cnn = calibration_curve(cnn_all_true, cnn_all_prob, n_bins=10, strategy="uniform")
frac_pos_hybrid, mean_pred_hybrid = calibration_curve(hybrid_all_true, hybrid_all_prob, n_bins=10, strategy="uniform")

brier_cnn = brier_score_loss(cnn_all_true, cnn_all_prob)
brier_hybrid = brier_score_loss(hybrid_all_true, hybrid_all_prob)

fig, ax = plt.subplots(figsize=(7, 7))
ax.plot([0, 1], [0, 1], linestyle="--", color="gray", label="Kalibrasi sempurna")
ax.plot(mean_pred_cnn, frac_pos_cnn, "o-", color="#339af0",
        label=f"CNN Baseline (Brier={brier_cnn:.3f})")
ax.plot(mean_pred_hybrid, frac_pos_hybrid, "s-", color="#e8590c",
        label=f"Hybrid CNN+Transformer (Brier={brier_hybrid:.3f})")
```

```

ax.set_xlabel("Probabilitas Prediksi Rata-rata (per bin)")
ax.set_ylabel("Proporsi Abnormal Aktual (per bin)")
ax.set_title("Calibration Curve - Agregat Prediksi 5-Fold")
ax.legend(loc="upper left")
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Brier Score CNN Baseline          : {brier_cnn:.4f}")
print(f"Brier Score CNN+Transformer Hybrid : {brier_hybrid:.4f}")
print("(Brier Score lebih rendah = probabilitas prediksi lebih dapat dipercaya)''')

cells.append(md("""## 14. Decision Curve Analysis (DCA)

Mengukur manfaat klinis (net benefit) model dibanding dua strategi ekstrem: **Treat All** (anggap semua pasien Abnormal) dan **Treat None** (anggap semua pasien Normal)

--- Grafik 7: Decision Curve Analysis (DCA) ---

def decision_curve_analysis(y_true, y_prob, thresholds):
    y_true = np.asarray(y_true)
    y_prob = np.asarray(y_prob)
    n = len(y_true)
    prevalence = y_true.mean()
    nb_model, nb_all = [], []
    for pt in thresholds:
        preds = (y_prob >= pt).astype(int)
        tp = np.sum((preds == 1) & (y_true == 1))
        fp = np.sum((preds == 1) & (y_true == 0))
        odds = pt / (1 - pt)
        nb_model.append(tp / n - fp / n * odds)
        nb_all.append(prevalence - (1 - prevalence) * odds)
    return np.array(nb_model), np.array(nb_all)

thresholds = np.linspace(0.01, 0.99, 99)
nb_cnn, nb_all_ref = decision_curve_analysis(cnn_all_true, cnn_all_prob, thresholds)
nb_hybrid, _ = decision_curve_analysis(hybrid_all_true, hybrid_all_prob, thresholds)

fig, ax = plt.subplots(figsize=(8, 6.5))
ax.plot(thresholds, nb_cnn, color="#339af0", linewidth=2, label="CNN Baseline")
ax.plot(thresholds, nb_hybrid, color="#e8590c", linewidth=2, label="CNN+Transformer Hybrid")
ax.plot(thresholds, nb_all_ref, color="#868e96", linestyle="--", linewidth=1.5, label="Treat All")
ax.axhline(0, color="black", linestyle=":", linewidth=1.5, label="Treat None")
ax.set_ylim(bottom=max(-0.1, min(nb_all_ref.min(), -0.05)))
ax.set_xlabel("Threshold Probability")
ax.set_ylabel("Net Benefit")
ax.set_title("Decision Curve Analysis - Agregat Prediksi 5-Fold")
ax.legend(loc="upper right")
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print("Interpretasi: kurva model yang berada di atas 'Treat All' dan 'Treat None' pada rentang")
print("threshold tertentu menunjukkan model memberi manfaat klinis nyata pada rentang tersebut.)''')

nb["cells"].extend(cells)
json.dump(nb, open(NB, "w", encoding="utf-8"), indent=1, ensure_ascii=False)
print("Appended", len(cells), "cells. Total now:", len(nb["cells"]))

```

3. build_pseudocode_doc.py

```
# -*- coding: utf-8 -*-
from docx import Document
from docx.shared import Pt, RGBColor, Cm
from docx.enum.text import WD_ALIGN_PARAGRAPH
from docx.enum.table import WD_TABLE_ALIGNMENT
from docx.oxml.ns import qn
from docx.oxml import OxmlElement

OUT_PATH = r"C:/Users/jhira/AppData/Local/Temp/claude/D--PROGRAM-ekg-cnn-transformers-/97040531-9273-4398-bca7-521b29e28e49"

NAVY = RGBColor(0x1B, 0x3A, 0x5C)
ACCENT = RGBColor(0x2E, 0x75, 0xB6)
GRAY = RGBColor(0x55, 0x55, 0x55)
ALGOBG = "F2F2F2"

doc = Document()
style = doc.styles["Normal"]
style.font.name = "Calibri"
style.font.size = Pt(11)
for i in range(1, 4):
    hstyle = doc.styles[f"Heading {i}"]
    hstyle.font.color.rgb = NAVY
    hstyle.font.name = "Calibri"

sec = doc.sections[0]
sec.page_width = Cm(21.0)
sec.page_height = Cm(29.7)
sec.left_margin = Cm(2.3)
sec.right_margin = Cm(2.3)
sec.top_margin = Cm(2.0)
sec.bottom_margin = Cm(2.0)

def set_cell_shading(cell, color_hex):
    shd = OxmlElement("w:shd")
    shd.set(qn("w:val"), "clear")
    shd.set(qn("w:color"), "auto")
    shd.set(qn("w:fill"), color_hex)
    cell._tc.get_or_add_tcPr().append(shd)

def algorithm_block(number, title, io_text, lines):
    """Render a pseudocode block as a shaded 1-cell table with monospace font."""
    table = doc.add_table(rows=1, cols=1)
    table.alignment = WD_TABLE_ALIGNMENT.CENTER
    cell = table.rows[0].cells[0]
    set_cell_shading(cell, ALGOBG)
    tcPr = cell._tc.get_or_add_tcPr()
    borders = OxmlElement("w:tcBorders")
    for edge in ("top", "left", "bottom", "right"):
        el = OxmlElement(f"w:{edge}")
        el.set(qn("w:val"), "single")
        el.set(qn("w:sz"), "10")
        el.set(qn("w:color"), "1B3A5C")
        borders.append(el)
    tcPr.append(borders)
    cell.text = ""
    p0 = cell.paragraphs[0]
    r0 = p0.add_run(f"Algorithm {number}: {title}")
```

```

r0.bold = True
r0.font.size = Pt(11.5)
r0.font.color.rgb = NAVY
r0.font.name = "Consolas"

p_io = cell.add_paragraph()
r_io = p_io.add_run(io_text)
r_io.italic = True
r_io.font.size = Pt(9.5)
r_io.font.name = "Consolas"

p_rule = cell.add_paragraph()
p_rule.paragraph_format.space_after = Pt(2)
pPr = p_rule._p.get_or_add_pPr()
pBdr = XmlElement("w:pBdr")
bottom = XmlElement("w:bottom")
bottom.set(qn("w:val"), "single")
bottom.set(qn("w:sz"), "6")
bottom.set(qn("w:space"), "1")
bottom.set(qn("w:color"), "1B3A5C")
pBdr.append(bottom)
pPr.append(pBdr)

for line_no, (indent, text) in enumerate(lines, start=1):
    p = cell.add_paragraph()
    p.paragraph_format.left_indent = Cm(0.4 + indent * 0.55)
    p.paragraph_format.space_after = Pt(1)
    r = p.add_run(f"{line_no:>2}: {text}")
    r.font.name = "Consolas"
    r.font.size = Pt(9.8)
    r.font.color.rgb = RGBColor(0x1A, 0x1A, 0x1A)

doc.add_paragraph()

def note_box(text, fill="FCE9C6", border="C98B1F"):
    table = doc.add_table(rows=1, cols=1)
    table.alignment = WD_TABLE_ALIGNMENT.CENTER
    cell = table.rows[0].cells[0]
    set_cell_shading(cell, fill)
    tcPr = cell._tc.get_or_add_tcPr()
    borders = XmlElement("w:tcBorders")
    for edge in ("top", "left", "bottom", "right"):
        el = XmlElement(f"w:{edge}")
        el.set(qn("w:val"), "single")
        el.set(qn("w:sz"), "8")
        el.set(qn("w:color"), border)
        borders.append(el)
    tcPr.append(borders)
    cell.text = ""
    p = cell.paragraphs[0]
    run = p.add_run(text)
    run.font.size = Pt(9.8)
    run.italic = True
    doc.add_paragraph()

# =====
# COVER
# =====
for _ in range(4):

```



```

doc.add_paragraph()

title = doc.add_paragraph()
title.alignment = WD_ALIGN_PARAGRAPH.CENTER
r = title.add_run("DOKUMEN PSEUDOCODE")
r.font.size = Pt(15)
r.font.color.rgb = GRAY
r.bold = True

title2 = doc.add_paragraph()
title2.alignment = WD_ALIGN_PARAGRAPH.CENTER
r2 = title2.add_run("Pipeline Preprocessing, Augmentasi, Arsitektur Model,\ndan Training – Klasifikasi Citra EKG")
r2.font.size = Pt(22)
r2.font.color.rgb = NAVY
r2.bold = True

sub = doc.add_paragraph()
sub.alignment = WD_ALIGN_PARAGRAPH.CENTER
rs = sub.add_run("CNN Baseline vs Hybrid CNN + Transformer")
rs.font.size = Pt(13)
rs.font.color.rgb = ACCENT

for _ in range(6):
    doc.add_paragraph()
for label, val in [
    ("Dokumen terkait", "reports/Visualisasi_Evaluasi_Tambahan_EKG.docx"),
    ("Notebook rujukan", "EKG_CNN_TRANSFORM.ipynb"),
    ("Jumlah algoritma", "5 blok pseudocode"),
]:
    p = doc.add_paragraph()
    p.alignment = WD_ALIGN_PARAGRAPH.CENTER
    r1 = p.add_run(f"{label}: ")
    r1.bold = True
    r1.font.size = Pt(11)
    r2_ = p.add_run(val)
    r2_.font.size = Pt(11)

doc.add_page_break()

# =====
# INTRO
# =====

doc.add_heading("Pendahuluan", level=1)
doc.add_paragraph(
    "Dokumen ini menyajikan representasi pseudocode formal dari lima komponen algoritmik utama "
    "proyek klasifikasi citra EKG: pipeline preprocessing, strategi augmentasi class-specific, "
    "arsitektur CNN Baseline, arsitektur Hybrid CNN+Transformer, dan prosedur training dengan "
    "5-fold stratified cross-validation. Notasi mengikuti konvensi umum pseudocode akademik "
    "(INPUT/OUTPUT eksplisit, indentasi untuk blok kontrol), diterjemahkan langsung dari "
    "implementasi Python/PyTorch aktual pada notebook EKG_CNN_TRANSFORM.ipynb."
)

doc.add_page_break()

# =====
# ALGORITHM 1: PREPROCESSING
# =====

doc.add_heading("Algorithm 1 – Preprocessing Pipeline", level=1)
doc.add_paragraph(
    "Tahap deterministik yang diterapkan sama pada seluruh 580 citra sebelum digunakan model."
)

```

```

algorithm_block(1, "Preprocess_Image", "INPUT: filepath (path citra mentah)\nOUTPUT: img_norm (array 224x224, float32, rent
    (0, "img_bgr &lt;- READ_IMAGE(filepath)",
    (0, "IF img_bgr is NULL THEN",
    (1, "RAISE ValueError(\"Gagal membaca gambar\")"),
    (0, "END IF"),
    (0, "img_rgb &lt;- CONVERT_COLOR(img_bgr, BGR_TO_RGB)",
    (0, "img_gray &lt;- CONVERT_COLOR(img_rgb, RGB_TO_GRAY)           // hilangkan color bias"),
    (0, "clahe &lt;- CREATE_CLAHE(clip_limit=2.0, tile_grid=(8,8))"),
    (0, "img_enhanced &lt;- clahe.APPLY(img_gray)           // perjelas gelombang EKG"),
    (0, "img_resized &lt;- RESIZE(img_enhanced, (224,224), method=INTER_AREA)",
    (0, "img_norm &lt;- CAST(img_resized, float32) / 255.0           // rentang [0,1]"),
    (0, "RETURN img_norm"),
])

# =====
# ALGORITHM 2: AUGMENTATION
# =====
doc.add_heading("Algorithm 2 – Class-Specific Augmentation", level=1)
doc.add_paragraph(
    "Diterapkan stokastik, hanya pada data training tiap fold. Kelas Abnormal menerima "
    "parameter 1.5x lebih agresif dibanding Normal."
)
algorithm_block(2, "Augment_Image", "INPUT: img (array 224x224 float32 [0,1]), label (\"Normal\" atau \"Abnormal\")\nOUTPUT
    (0, "mult &lt;- IF label == \"Abnormal\" THEN 1.5 ELSE 1.0"),
    (0, "p &lt;- SCALE_PARAMS(BASE_AUG_PARAMS, mult) // rotasi, shift, zoom, brightness, noise, blur, apply_prob"),
    (0, "out &lt;- COPY(img)",
    (0, "IF RANDOM() &lt; p.apply_prob THEN out &lt;- ROTATE(out, ±p.rotation_deg)           END IF"),
    (0, "IF RANDOM() &lt; p.apply_prob THEN out &lt;- SHIFT(out, ±p.shift_pct)           END IF"),
    (0, "IF RANDOM() &lt; p.apply_prob THEN out &lt;- ZOOM(out, ±p.zoom_pct)           END IF"),
    (0, "IF RANDOM() &lt; p.apply_prob THEN out &lt;- ADJUST_BRIGHTNESS(out, p.brightness_range) END IF"),
    (0, "IF RANDOM() &lt; p.apply_prob THEN out &lt;- ADD_GAUSSIAN_NOISE(out, mu=0, sigma=p.noise_sigma) END IF"),
    (0, "IF RANDOM() &lt; p.apply_prob THEN out &lt;- GAUSSIAN_BLUR(out, kernel=3, sigma=p.blur_sigma_range) END IF"),
    (0, "RETURN CLIP(out, 0, 1)"),
])

doc.add_page_break()

# =====
# ALGORITHM 3: CNN BASELINE
# =====
doc.add_heading("Algorithm 3 – CNN Baseline Forward Pass (ResNet18 Transfer Learning)", level=1)
algorithm_block(3, "CNN_Baseline_Forward", "INPUT: x (tensor B x 3 x 224 x 224, hasil 3ch-duplikasi + normalisasi ImageNet)
    (0, "backbone &lt;- LOAD_PRETRAINED(ResNet18, weights=ImageNet1K)",
    (0, "FREEZE(backbone.conv1, backbone.bn1, backbone.layer1, backbone.layer2, backbone.layer3)",
    (0, "UNFREEZE(backbone.layer4)           // hanya blok terakhir dilatih"),
    (0, "backbone.fc &lt;- Linear(in_features=512, out_features=2) // classifier head baru"),
    (0, ""),
    (0, "// forward pass:"),
    (0, "feat &lt;- backbone.conv1(x) -&gt; backbone.bn1 -&gt; ReLU -&gt; MaxPool",
    (0, "feat &lt;- backbone.layer1(feat) -&gt; layer2(feat) -&gt; layer3(feat) -&gt; layer4(feat)",
    (0, "pooled &lt;- GlobalAveragePool(feat)           // (B, 512)",
    (0, "logits &lt;- backbone.fc(pooled)           // (B, 2)",
    (0, "RETURN logits"),
])

# =====
# ALGORITHM 4: HYBRID CNN+TRANSFORMER
# =====

```

```

doc.add_heading("Algorithm 4 – Hybrid CNN+Transformer Forward Pass", level=1)
algorithm_block(4, "Hybrid_CNN_Transformer_Forward", "INPUT: x (tensor B x 3 x 224 x 224)\nOUTPUT: logits (tensor B x 2)",
    (0, "backbone &lt;- LOAD_PRETRAINED(ResNet18, weights=ImageNet1K) MINUS {avgpool, fc}"),
    (0, "FREEZE(backbone layers KECUALI layer4)"),
    (0, "cls_token &lt;- LEARNABLE_PARAM(shape=(1,1,512))"),
    (0, "pos_embed &lt;- LEARNABLE_PARAM(shape=(1,50,512))      // 49 patch + 1 CLS"),
    (0, "transformer &lt;- TransformerEncoder(layers=2, heads=8, dim_ff=1024, dropout=0.1)"),
    (0, "classifier &lt;- Linear(in_features=512, out_features=2)"),
    (0, ""),
    (0, "// forward pass:"),
    (0, "feat_map &lt;- backbone(x)                                // (B, 512, 7, 7)"),
    (0, "tokens &lt;- FLATTEN_SPATIAL(feat_map)                    // (B, 49, 512)"),
    (0, "tokens &lt;- CONCAT(EXPAND(cls_token, B), tokens)        // (B, 50, 512)"),
    (0, "tokens &lt;- tokens + pos_embed"),
    (0, "encoded &lt;- transformer(tokens)                        // (B, 50, 512), self-attention 8-head"),
    (0, "cls_out &lt;- LayerNorm(encoded[:, 0])                  // ambil token CLS"),
    (0, "logits &lt;- classifier(cls_out)                        // (B, 2)"),
    (0, "RETURN logits"),
])

doc.add_page_break()

# =====
# ALGORITHM 5: TRAINING
# =====

doc.add_heading("Algorithm 5 – Training dengan 5-Fold Stratified Cross-Validation", level=1)
doc.add_paragraph(
    "Prosedur identik dijalankan untuk kedua model (CNN Baseline dan Hybrid), dengan hyperparameter "
    "tetap (tanpa tuning) agar perbandingan adil."
)

algorithm_block(5, "Train_With_5Fold_CV", "INPUT: D (dataset 580 citra + label), model_builder (fungsi konstruksi model)\nOUTPUT: results (dict)",
    (0, "folds &lt;- StratifiedKFold(D, n_splits=5, shuffle=True, seed=42)"),
    (0, "results &lt;- []"),
    (0, "FOR fold_idx, (train_idx, test_idx) IN ENUMERATE(folds) DO"),
    (1, "class_weight &lt;- COMPUTE_INVERSE_FREQ_WEIGHT(D.labels[train_idx])"),
    (1, "train_loader &lt;- DataLoader(D[train_idx], augment=TRUE, batch_size=16, shuffle=True)"),
    (1, "test_loader &lt;- DataLoader(D[test_idx], augment=FALSE, batch_size=16, shuffle=False)"),
    (1, "model &lt;- model_builder()"),
    (1, "optimizer &lt;- Adam(model.trainable_params, lr=1e-4)"),
    (1, "loss_fn &lt;- CrossEntropyLoss(weight=class_weight)"),
    (1, ""),
    (1, "FOR epoch IN 1..12 DO"),
    (2, "FOR (images, labels) IN train_loader DO"),
    (3, "logits &lt;- model(images)"),
    (3, "loss &lt;- loss_fn(logits, labels)"),
    (3, "BACKPROPAGATE(loss); optimizer.STEP(); optimizer.ZERO_GRAD()"),
    (2, "END FOR"),
    (1, "END FOR"),
    (1, ""),
    (1, "y_true, y_pred, y_prob &lt;- EVALUATE(model, test_loader)"),
    (1, "metrics &lt;- COMPUTE(Accuracy, Precision, Recall, F1, AUC | y_true, y_pred, y_prob)"),
    (1, "results.APPEND(metrics)"),
    (0, "END FOR"),
    (0, "RETURN results"),
])

doc.add_paragraph(

```

```
note_box(  
    "Catatan: prosedur ini dijalankan dua kali secara terpisah pada notebook – sekali dengan "  
    "model_builder = CNN_Baseline_Forward (Algorithm 3), sekali dengan model_builder = "  
    "Hybrid_CNN_Transformer_Forward (Algorithm 4) – menghasilkan dua set metrics_per_fold yang "  
    "diperbandingkan pada laporan analisis utama."  
)  
doc.save(OUT_PATH)  
print("Saved:", OUT_PATH)
```

4. build_extra_viz_doc.py

```
# -*- coding: utf-8 -*-
import os
from docx import Document
from docx.shared import Inches, Pt, RGBColor, Cm
from docx.enum.text import WD_ALIGN_PARAGRAPH
from docx.enum.table import WD_TABLE_ALIGNMENT
from docx.oxml.ns import qn
from docx.oxml import OxmlElement

IMG_DIR = r"C:/Users/jhira/AppData/Local/Temp/claude/D--PROGRAM-ekg-cnn-transformers-/4d03bc44-65f2-4f8c-a560-ca6073f82ece/"
OUT_PATH = r"C:/Users/jhira/AppData/Local/Temp/claude/D--PROGRAM-ekg-cnn-transformers-/4d03bc44-65f2-4f8c-a560-ca6073f82ece/"

IMG = lambda name: os.path.join(IMG_DIR, name)

NAVY = RGBColor(0x1B, 0x3A, 0x5C)
ACCENT = RGBColor(0x2E, 0x75, 0xB6)
GRAY = RGBColor(0x55, 0x55, 0x55)

doc = Document()
style = doc.styles["Normal"]
style.font.name = "Calibri"
style.font.size = Pt(11)
for i in range(1, 4):
    hstyle = doc.styles[f"Heading {i}"]
    hstyle.font.color.rgb = NAVY
    hstyle.font.name = "Calibri"

sec = doc.sections[0]
sec.page_width = Cm(21.0)
sec.page_height = Cm(29.7)
sec.left_margin = Cm(2.3)
sec.right_margin = Cm(2.3)
sec.top_margin = Cm(2.0)
sec.bottom_margin = Cm(2.0)

def set_cell_shading(cell, color_hex):
    shd = OxmlElement("w:shd")
    shd.set(qn("w:val"), "clear")
    shd.set(qn("w:color"), "auto")
    shd.set(qn("w:fill"), color_hex)
    cell._tc.get_or_add_tcPr().append(shd)

def add_table(headers, rows, widths_cm=None, header_fill="1B3A5C", font_size=9.5):
    table = doc.add_table(rows=1, cols=len(headers))
    table.style = "Table Grid"
    table.alignment = WD_TABLE_ALIGNMENT.CENTER
    hdr_cells = table.rows[0].cells
    for i, h in enumerate(headers):
        hdr_cells[i].text = ""
        p = hdr_cells[i].paragraphs[0]
        run = p.add_run(h)
        run.bold = True
        run.font.color.rgb = RGBColor(0xFF, 0xFF, 0xFF)
        run.font.size = Pt(font_size)
        p.alignment = WD_ALIGN_PARAGRAPH.CENTER
        set_cell_shading(hdr_cells[i], header_fill)
    for row_vals in rows:
        cells = table.add_row().cells
```

```

        for i, val in enumerate(row_vals):
            cells[i].text = ""
            p = cells[i].paragraphs[0]
            run = p.add_run(str(val))
            run.font.size = Pt(font_size)
            p.alignment = WD_ALIGN_PARAGRAPH.CENTER

    if widths_cm:
        for row_ in table.rows:
            for i, w in enumerate(widths_cm):
                row_.cells[i].width = Cm(w)
    doc.add_paragraph()

def add_figure(img_name, caption, width_in=6.3):
    p = doc.add_paragraph()
    p.alignment = WD_ALIGN_PARAGRAPH.CENTER
    run = p.add_run()
    run.add_picture(IMG(img_name), width=Inches(width_in))
    cap = doc.add_paragraph()
    cap.alignment = WD_ALIGN_PARAGRAPH.CENTER
    cap_run = cap.add_run(caption)
    cap_run.italic = True
    cap_run.font.size = Pt(9.5)
    cap_run.font.color.rgb = GRAY
    doc.add_paragraph()

def add_bullets(items):
    for it in items:
        doc.add_paragraph(it, style="List Bullet")

# =====
# COVER
# =====

for _ in range(4):
    doc.add_paragraph()

title = doc.add_paragraph()
title.alignment = WD_ALIGN_PARAGRAPH.CENTER
r = title.add_run("VISUALISASI EVALUASI TAMBAHAN")
r.font.size = Pt(15)
r.font.color.rgb = GRAY
r.bold = True

title2 = doc.add_paragraph()
title2.alignment = WD_ALIGN_PARAGRAPH.CENTER
r2 = title2.add_run("Precision-Recall per Fold, Calibration Curve,\nDecision Curve Analysis, dan ROC Curve")
r2.font.size = Pt(22)
r2.font.color.rgb = NAVY
r2.bold = True

sub = doc.add_paragraph()
sub.alignment = WD_ALIGN_PARAGRAPH.CENTER
rs = sub.add_run("Klasifikasi Citra EKG – CNN Baseline vs Hybrid CNN + Transformer")
rs.font.size = Pt(13)
rs.font.color.rgb = ACCENT

for _ in range(6):
    doc.add_paragraph()

for label, val in [
    ("Dokumen terkait", "reports/Pseudocode_EKG_CNN_Transformer.docx"),
    ("Notebook rujukan", "EKG_CNN_TRANSFORM.ipynb"),

```

```

("Basis evaluasi", "5-Fold Stratified Cross-Validation, agregat 580 sampel"),
]:
p = doc.add_paragraph()
p.alignment = WD_ALIGN_PARAGRAPH.CENTER
r1 = p.add_run(f"{label}: ")
r1.bold = True
r1.font.size = Pt(11)
r2_ = p.add_run(val)
r2_.font.size = Pt(11)

doc.add_page_break()

# =====
# INTRO
# =====
doc.add_heading("Pendahuluan", level=1)
doc.add_paragraph(
    "Dokumen ini memuat empat visualisasi evaluasi tambahan yang melengkapi laporan analisis utama "
    "(reports/Laporan_Analisis_Klasifikasi_EKG_CNN_Transformer.docx): grafik trade-off "
    "precision-recall antar fold, calibration curve, Decision Curve Analysis (DCA), dan ROC curve. "
    "Seluruh grafik dihasilkan dari prediksi agregat 5-fold cross-validation pada 580 sampel citra "
    "EKG, membandingkan model CNN Baseline (ResNet18 transfer learning) dengan model Hybrid "
    "CNN+Transformer."
)
doc.add_page_break()

# =====
# 1. PRECISION VS RECALL PER FOLD
# =====
doc.add_heading("1. Precision vs Recall per Fold (5-Fold CV)", level=1)
doc.add_paragraph(
    "Grafik garis yang menghubungkan titik precision-recall tiap fold secara berurutan (Fold 1 -& "
    "Fold 5), memvisualisasikan langsung trade-off precision-recall serta konsistensi (atau "
    "variabilitas) performa antar fold pada kedua model."
)
add_figure("Grafik_5_Precision.png", "Gambar 1. Precision (Abnormal) vs Recall (Abnormal) per fold, CNN Baseline vs Hybrid
doc.add_paragraph(
    "Pengamatan: titik-titik CNN Baseline berkerumun di area precision tinggi (0.90-1.00) namun "
    "recall lebih rendah (0.49-0.68) – pola model yang konservatif. Titik-titik Hybrid tersebar "
    "lebih ke kanan (recall 0.53-0.76) dengan precision sedikit lebih rendah (0.80-0.96), "
    "mengonfirmasi temuan pada laporan utama: Hybrid mengambil sikap lebih berani dalam mendeteksi "
    "kasus Abnormal dengan konsekuensi presisi yang sedikit menurun."
)
doc.add_page_break()

# =====
# 2. CALIBRATION CURVE
# =====
doc.add_heading("2. Calibration Curve (Reliability Diagram)", level=1)
doc.add_paragraph(
    "Calibration curve menguji apakah nilai probabilitas yang dikeluarkan model dapat dipercaya "
    "secara harfiah – bukan sekadar dipakai untuk mengurutkan (ranking) kelas seperti pada AUC. "
    "Sumbu-X menunjukkan probabilitas prediksi rata-rata dalam tiap bin (10 bin, strategi uniform); "
    "sumbu-Y menunjukkan proporsi aktual kelas Abnormal pada bin tersebut. Kalibrasi sempurna "
    "ditunjukkan oleh garis diagonal putus-putus – jika model menyatakan \"70% yakin Abnormal\", "
    "idealnya benar-benar 70% dari kasus pada bin itu adalah Abnormal."
)
add_figure("Grafik_6_Calibration.png", "Gambar 2. Calibration curve (reliability diagram), agregat prediksi 5-fold.")

```

```

add_table(
    ["Model", "Brier Score"],
    [
        ["CNN Baseline", "0.1382"],
        ["CNN+Transformer Hybrid", "0.1315"],
    ],
    widths_cm=[8, 5],
)
doc.add_paragraph(
    "Brier Score (rata-rata kuadrat selisih antara probabilitas prediksi dan label aktual, "
    "semakin rendah semakin baik) menunjukkan Hybrid memiliki kalibrasi probabilitas yang sedikit "
    "lebih baik dibanding CNN Baseline. Kedua kurva menunjukkan fluktuasi di sekitar garis diagonal "
    "yang wajar mengingat jumlah sampel per bin relatif kecil (~58 sampel/bin dari 580 total), "
    "sehingga estimasi proporsi per bin cenderung noisy – bukan indikasi masalah kalibrasi yang "
    "serius pada kedua model."
)
doc.add_page_break()
# =====
# 3. DECISION CURVE ANALYSIS
# =====
doc.add_heading("3. Decision Curve Analysis (DCA)", level=1)
doc.add_paragraph(
    "DCA (Vickers & Elkin, 2006) mengukur manfaat klinis (net benefit) suatu model prediksi "
    "dibanding dua strategi ekstrem: \"Treat All\" (anggap seluruh pasien Abnormal, rujuk semua) "
    "dan \"Treat None\" (anggap seluruh pasien Normal, tidak merujuk siapa pun). Sumbu-X adalah "
    "threshold probability – ambang keputusan klinis (\"jika model yakin > X%, maka rujuk\"). "
    "Kurva model yang berada di atas kedua strategi ekstrem pada rentang threshold tertentu "
    "menunjukkan model memberi nilai tambah nyata dibanding heuristik sederhana pada rentang "
    "tersebut."
)
add_figure("Grafik_7_Decision_Curve.png", "Gambar 3. Decision Curve Analysis, agregat prediksi 5-fold.")
doc.add_paragraph(
    "Pengamatan: kedua model (CNN Baseline dan Hybrid) konsisten berada di atas kurva \"Treat All\" "
    "(yang menurun tajam dan negatif setelah threshold ~0.35) maupun \"Treat None\" (net benefit=0) "
    "di hampir seluruh rentang threshold yang relevan secara klinis (0.05-0.90). Ini menunjukkan "
    "kedua model memberi manfaat keputusan klinis yang nyata dibanding strategi naif \"rujuk semua\" "
    "atau \"jangan rujuk siapa pun\", pada rentang ambang keputusan yang luas."
)
doc.add_page_break()
# =====
# 4. ROC CURVE
# =====
doc.add_heading("4. ROC Curve", level=1)
doc.add_paragraph(
    "ROC (Receiver Operating Characteristic) curve memplot True Positive Rate (Recall/Sensitivity) "
    "terhadap False Positive Rate (1-Specificity) pada seluruh kemungkinan threshold klasifikasi. "
    "AUC (Area Under Curve) merangkum kemampuan model memisahkan kedua kelas secara keseluruhan, "
    "tidak terikat pada satu threshold tertentu – 0.5 berarti setara tebakan acak, 1.0 berarti "
    "pemisahan sempurna."
)
add_figure("Grafik_3_ROC_Curve.png", "Gambar 4. ROC curve, agregat prediksi 5-fold, CNN Baseline vs Hybrid CNN+Transformer.")
doc.add_paragraph(
    "AUC CNN Baseline = 0.870, AUC Hybrid CNN+Transformer = 0.893. Hybrid unggul +0.023 pada "
    "metrik yang tidak bergantung threshold ini, mengonfirmasi temuan pada laporan analisis utama "

```



```

        "bahwa penambahan komponen Transformer memberi peningkatan kemampuan diskriminasi kelas yang "
        "konsisten."
    )
doc.add_page_break()

# =====
# RINGKASAN
# =====
doc.add_heading("Ringkasan", level=1)
add_table(
    ["Visualisasi", "Temuan Utama"],
    [
        ["Precision-Recall per Fold", "CNN konsisten precision tinggi/recall rendah; Hybrid lebih seimbang dengan recall le
        ["Calibration Curve", "Hybrid Brier Score sedikit lebih baik (0.132 vs 0.138); kedua model cukup terkalibrasi"],
        ["Decision Curve Analysis", "Kedua model memberi net benefit positif di atas Treat All/Treat None pada rentang thre
        ["ROC Curve", "Hybrid AUC lebih tinggi (0.893 vs 0.870), konsisten dengan temuan laporan utama"],
    ],
    widths_cm=[5, 10],
    font_size=9,
)
doc.save(OUT_PATH)
print("Saved:", OUT_PATH)

```